

EDUCATION

University of Michigan (USA)

Sept. 2020 – Jun. 2026

Ph.D. in Physics

University of Science and Technology of China (China)

Sept. 2015 – Jul. 2019

B.S. in Physics

RESEARCH EXPERIENCE

Research Assistant

Sept. 2020 – Jun. 2026

University of Michigan, MI, USA

- Controlled Analytic Continuation from Matsubara data
 - * Presented a systematically improvable approach for numerical analytic continuation
 - * Developed a compact representation to efficiently compress Matsubara data with minimal degrees of freedom
 - * Created the [PronyAC](#) and [MiniPole](#) libraries for analytic continuation and beyond
- Real-frequency fitting for spectral functions
 - * Developed a method to achieve high-precision fitting of the real-frequency spectral function using a minimal number of complex poles
 - * Resolved artificial parameter dependencies and other limitations inherent in existing methodologies
- Non-crossing and one-crossing approximations for quantum impurity models
 - * Developed C++ code for non-crossing and one-crossing approximations
 - * Utilized various schemes such as block-diagonalization and non-uniform meshes to speed up the computation
- Inchworm algorithm for multi-orbital steady-state impurity models
 - * Studied inchworm algorithm for non-equilibrium systems
 - * Participated in the development of the InchMOSS project

Research Assistant

Jul. 2017 – Jul. 2020

University of Science and Technology of China, AH, China

- Loop-Cluster algorithm for q -state Potts model
 - * Presented the solution to couple different representations of the q -state Potts model
 - * Formulated a novel Monte Carlo algorithm: Loop-Cluster (LC) algorithm
 - * Proposed an efficient method to carry out simulations
- Graphical representations and worm algorithms for the $O(N)$ spin model
 - * Presented a variety of graphical representations for the classical $O(N)$ spin model
 - * Formulated corresponding worm algorithms to perform simulations
 - * Explored the dynamic properties of these algorithms

PUBLICATIONS

- **“Multi-orbital dynamical mean-field theory with a complex-time solver”**,
by Yang Yu, **Lei Zhang**, Emanuel Gull, Xiaodong Cao and Xinyang Dong,
available on 29 December 2025 in [arXiv:2512.23237](#).
- **“Compact representation and long-time extrapolation of real-time data for quantum systems using the ESPRIT algorithm”**,
by André Erpenbeck, Yuanran Zhu, Yang Yu, **Lei Zhang**, Richard Gerum, Olga Goulko, Chao Yang, Guy Cohen and Emanuel Gull,
published on 16 March 2026 in [Phys. Rev. B 113, 115129 \(2026\)](#).
- **“Minimal pole representation for spectral functions”**,
by **Lei Zhang**, André Erpenbeck, Yang Yu and Emanuel Gull,
published on 3 June 2025 in [J. Chem. Phys. 162, 214111 \(2025\)](#).
- **“Minimal pole representation and analytic continuation of matrix-valued correlation functions”**,
by **Lei Zhang**, Yang Yu and Emanuel Gull,
published on 12 December 2024 in [Phys. Rev. B 110, 235131 \(2024\)](#) [Editors’ Suggestion].
- **“Green/WeakCoupling: Implementation of fully self-consistent finite-temperature many-body perturbation theory for molecules and solids”**,
by Sergei Isakov, Chia-Nan Yeh, Pavel Pokhilko, Yang Yu, **Lei Zhang**, Gaurav Harsha, Vibin Abraham, Ming Wen, Munkhorgil Wang, Jacob Adamski, Tianran Chen, Emanuel Gull and Dominika Zgid, published in [Comput. Phys. Commun. 306, 109380 \(2025\)](#).
- **“Steady-state properties of multi-orbital systems using quantum Monte Carlo”**,
by Andre Erpenbeck, Thomas Blommel, **Lei Zhang**, Wei-Ting Lin, Guy Cohen and Emanuel Gull, published on 4 September 2024 in [J. Chem. Phys. 161, 094104 \(2024\)](#).
- **“Feynman diagrammatics based on discrete pole representations: A path to renormalized perturbation theories”**,
by Daria Gazizova, **Lei Zhang**, Emanuel Gull and J. P. F. LeBlanc,
published on 27 August 2024 in [Phys. Rev. B 110, 075158 \(2024\)](#).
- **“Minimal Pole Representation and Controlled Analytic Continuation of Matsubara Response Functions”**,
by **Lei Zhang** and Emanuel Gull,
published on 24 July 2024 in [Phys. Rev. B 110, 035154 \(2024\)](#).
- **“Tensor train continuous time solver for quantum impurity models”**,
by A. Erpenbeck, W.-T. Lin, T. Blommel, **L. Zhang**, S. Isakov, L. Bernheimer, Y. Núñez-Fernández, G. Cohen, O. Parcollet, X. Waintal and E. Gull,
published on 26 June 2023 in [Phys. Rev. B 107, 245135 \(2023\)](#).
- **“Graphical Representations and Worm Algorithms for the $O(N)$ Spin Model”**,
by Longxiang Liu*, **Lei Zhang***, Xiaojun Tan and Youjin Deng,
published on 10 November 2023 in [Commun. Theor. Phys. 75, 115702 \(2023\)](#),
(*: equal contribution).
- **“Loop-Cluster Coupling and Algorithm for Classical Statistical Models”**,
by **Lei Zhang**, Manon Michel, Eren M. Elçi and Youjin Deng,
published on 12 November 2020 in [Phys. Rev. Lett. 125, 200603 \(2020\)](#).

PRESENTATIONS

Oral Presentations

- Complex Pole Representation in Frequency and Its Applications to Many-Body Problems: APS Global Physics Summit 2026 (Denver)
- Controlled analytic continuation of Matsubara correlation functions using minimal information principle: invited talk at USTC, 2025
- Minimal pole method for spectral functions: invited talk at USTC, 2025
- Controlled analytic continuation of Matsubara correlation functions using minimal pole representation: APS Global Physics Summit 2025 (Anaheim) (Condensed Matter community)
- A fresh look at the analytic continuation problem: insights from the minimum information principle and from Nevanlinna theory: APS Global Physics Summit 2025 (Anaheim) (Lattice QCD community)
- Loop-Cluster Coupling and Algorithm for Classical Statistical Models: APS March Meeting 2021 (Online) and 2022 (Chicago)

Poster Presentations

- Controlled analytic continuation of Matsubara correlation functions using minimal pole representation:
School on Electron Correlations and Topology, Houston, TX, USA, 2025
Electronic Structure Workshop, El Paso, TX, USA, 2025
Simons Foundation Superconductivity Summer School, Minneapolis, MN, USA, 2025
- Application of the Prony Method in Analytic Continuation:
Autumn School on Correlated Electrons, Jülich, Germany, 2023
MQC Entanglement Conference, Lansing, MI, USA, 2024

SELECTED AWARDS

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| • Rackham One-Term Dissertation Fellowship | 2025 |
| • MICDE Fellowship | 2024 |
| • Outstanding Graduate of Anhui Province | 2019 |
| • Outstanding Graduate of USTC | 2019 |
| • Special Scholarship of SINANO, Chinese Academy of Sciences | 2018 |
| • National Encouragement Scholarship | 2017 |
| • First Prize in National College Students' Mathematics Competition | 2016 & 2017 |
| • Yan Jici Scholarship, Chinese Academy of Sciences | 2015 – 2019 |

TEACHING EXPERIENCE

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|---|---------------------------|
| • Physics 506 (Electromagnetism II), graduate course | Winter 2022 & Winter 2023 |
| • Physics 505 (Electromagnetism I), graduate course | Fall 2022 |
| • Physics 151 (Life Sciences Lab I), undergraduate course | Fall 2021 |
| • Physics 136 (Life Sciences Lab I), undergraduate course | Fall 2020 & Winter 2021 |

TECHNICAL SKILLS

Programming Languages: Python, C++, Fortran, MATLAB, Julia, Java, Mathematica

Libraries and Tools: NumPy, Matplotlib, Mpmath, Eigen, CMake, Git, $\LaTeX$, TikZ, Gnuplot